Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/30/2019
PI Name:	Govind, Shubha Ph.D.		
Project Title:	Does Spaceflight Alter the Virulence of a Natural Parasite of Drosophila?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	sgovind@ccny.cuny.edu	**Fax:**	FY 212-650-8585
PI Organization Type:	UNIVERSITY	**Phone:**	212-650-8476
Organization Name:	City College of New York		
PI Address 1:	Biology Department		
PI Address 2:	160 Convent Ave		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10031-9101	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/31/2021 per NSSC information (Ed., 8/4/2020) NOTE: End date changed to 10/31/2020 per F. Hernandez/ARC (Ed., 6/20/19) NOTE: End date changed to 10/31/2019 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 10/31/2018 per F. Hernandez/ARC (Ed., 9/21/17) NOTE: End date changed to 10/31/2017 per NSSC information (Ed., 8/3/16)		
Key Personnel Changes/Previous PI:	September 2019 report: (1) PhD student Mr. Waleed Khalid went on leave of absence Fall 2018 and Spring 2019 semesters. (2) Increased Jennifer Chou's (technical assistant) time commitment. Her appointment is terminating Aug 31, 2019 due to limited funds. (3) PhD student Brian Wey was supported full time on the grant but will be teaching in the coming academic year due to limited funds on the project.		
COI Name (Institution):	Bhattacharya, Sharmila Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX15AB42G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The seminal discovery of Toll and Toll-NF-KB signaling as a central mediator of innate immune responses in the model insect Drosophila catalyzed similar discoveries in other organisms including innate immunity in humans. These findings led to an appreciation of the fundamental roles of innate immunity in human immune deficiency on the one hand, and immune health of insects on the other. In Drosophila larvae, primed, but quiescent hematopoietic progenitors are called into action from an organ with functional similarities to mammalian bone marrow. Blood cells signal the fat body (equivalent to the mammalian liver) to activate the expression of antimicrobial peptide genes (humoral immunity). Antimicrobial peptides efficiently combat deadly bacteria and fungi. The Leptopilina wasps are natural metazoan enemies of Drosophila and their attack kills fruit flies. Wasp attack (e.g., L. boulardi) activates both the humoral and cellular immune arms of larval hosts, each being dependent on Toll-NFKB signaling. Previous NASA-funded studies of Drosophila have shown that stresses in the space environment compromise hematopoietic development and modulate larval and adult innate immunity. We hypothesize that stimuli from the space environment will also alter the virulence of the parasitic wasps, and therefore their success on the fly hosts cultured in space. In Drosophila-wasp co-culture experiments using the NASA Fruit Fly Lab hardware over 30-days, we propose to (1) examine whether the fly or the wasp partner wins the arms race in space; (2) examine the effects of spaceflight on the anti-parasite response in fruit fly hosts; and (3) examine if virulence of "space-cultured" wasps is different from counterparts grown on Earth. While both L. boulardi and L. heterotoma parasites always succeed on the wild type, only the latter wasp species produces unique cell-killing activities residing in its virus-like particles. It also suppresses humoral immunity completely. We will examine if changes in the expression of VLP (virus-like particles) genes underlies changes in parasite virulence affected by spaceflight. Host defense and virulence will be inferred from a suite of other well-established assays of fly/host survival and measures of immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Parasitic wasps of Drosophila are a powerful and increasingly well-characterized model for understanding longstanding questions about natural host/parasite interactions, allowing molecular genetic examination of both innate immune mechanisms in fly hosts and virulence mechanisms of their wasp parasites. Distant relatives of these parasitic wasps are used for biocontrol of agricultural pests worldwide. The "Earth benefits" impact of our research over the last yeast has been as follows: (a) Ground and space whole body transcriptomes of L. heterotoma obtained. Forty putative venom genes whose transcription is affected in either condition identified. These genes are candidates for ongoing analysis. Transcriptomic data have been submitted to NCBI (National Center for Biotechnology Information) for release to the public upon publication of results. The ground transcriptome is part of a publication that is under review. The space transcriptome and comparison is in manuscript in preparation. (b) Identification and isolation of viable mutants of L. heterotoma with golden wing color. Gray wing is the normal color. (c) Over 10 students directly trained in the lab over last year. One technicians trained and contributed to the project. Six Federal Work Study students trained contributed to the project. Collaborations developed that resulted in coauthored manuscript. (d) One collaborative paper published, two other papers revised and resubmitted and are currently in review. Several oral and poster presentations made to disseminate research results.
Task Progress:	Space exploration presents significant challenges to human physiology, affecting all biological systems. Effects of spaceflight on human immunity are of particular interest, especially in the context of exploration-class missions. A number of studies have focused on immune dysfunction in space, but there remain significant gaps in our understanding of the specific effects of spaceflight and microgravity on the innate immune system at the cellular and molecular-genetic levels. Drosophila melanogaster is a simple and genetically accessible model system, deficient of most hallmarks of adaptive immunity. The seminal discoveries of the Toll-like receptors and Toll-NF-kB signaling as central mediators of innate immune responses in mammals were catalyzed by earlier genetic studies in Drosophila. More recently, studies have uncovered the effects of spaceflight on immune gene expression, blood cell dysfunction, and host competence against microbial infections. To better understand the effects of spaceflight on host-parasite interaction, natural parasitic wasps of Drosophila were sent to the International Space Station (ISS) in April 2018. The parasites attack the larval stages of fruit flies. After laying eggs in fly larvae, they hatch and eat the developing fly host, and emerge from the pupal case in about 20-25 days. The overarching goals of the study were two-fold: First to examine if the success of the fly/wasp partner is altered in space (who wins the "arms race"); and second, to determine, if parasites reared in space have the same virulence (or reduced virulence) as their counterparts on Earth. Over the last year, we have focused efforts on virulence and staining assays for both wasps and genomic and transcriptomic analyses of Leptopilina heterotoma. We have isolated visible mutants of this wasp, likely induced in the space environment. Studies with L. boulardi are ongoing.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Abstracts for Journals and Proceedings	Govind S, Chou J, Chen J, Heavner ME, Khalid W, Ramroop J, Razzak Z, Wey B, Blazevic E, Gresser A, Lera M, Martin K, Torres M, Bhattacharya S. "FFL-03: Successful development of Drosophila parasites aboard the International Space Station." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Abstract # 248. , Nov-2018

Abstracts for Journals and Proceedings	Govind S. "Drosophila immunity and parasitoid virulence factors: the host parasitoid interface." 2018 Joint ESA, ESC, and ESBC Annual Meeting--Crossing Borders: Entomology in a Changing World, Vancouver, British Columbia, November 11-14, 2018. Member Symposium: Evolutionary Ecology of Drosophila-parasitoid Food Webs: From Molecules to Ecosystems, Vancouver, British Columbia, November 11-14, 2018. , Nov-2018
Articles in Peer-reviewed Journals	Panettieri S, Paddibhatla I, Chou J, Rajwani R, Moore R, Goncharuk T, John G, Govind S. "Discovery of aspirin-triggered eicosanoid-like mediators in a Drosophila metainflammation-blood tumor model." J Cell Sci. 2019 Oct 28;133(5):jcs236141. https://doi.org/10.1242/jcs.236141 ; PMID: 31562189; PMCID: PMC7033724 , Oct-2019
Articles in Peer-reviewed Journals	Scarpati M, Qi Y, Govind S, Singh S. "A combined computational strategy of sequence and structural analysis predicts the existence of a functional eicosanoid pathway in Drosophila melanogaster." PLoS One. 2019 Feb 12;14(2):e0211897. https://doi.org/10.1371/journal.pone.0211897 ; PubMed PMID: 30753230; PubMed Central PMCID: PMC6372189 , Feb-2019